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Magnetostratigraphy and paleontology of Aït Kandoula Basin (High Atlas, Morocco) and the African–European late Miocene terrestrial fauna exchanges

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Abstract

A magnetostratigraphic study has been carried out on a middle Miocene to upper Pliocene lacustrine sedimentary deposit in the central part of the Aït Kandoula basin, which contains micromammal faunas and is situated in the southern High Atlas (Morocco). In total, 113 samples were subjected to paleomagnetic analysis: 60 out of the 113 studied samples representing 52 different stratigraphic levels yielded a paleomagnetic direction and at least the polarity could be recognized in 42 specimens. Eleven specimens were submitted to AF demagnetization. The mean direction for normal-polarity samples was $D = 349.4$, $I = 50.7$ ($N = 36$, $\alpha_{95} = 4.5$, $k = 27$) and for reversed polarity samples $D = 191.2$, $I = -53.4$ ($N = 16$, $\alpha_{95} = 12.32$, $k = 9$). These results yielded a polarity sequence which we interpret as spanning from Chron C5n.2n to the beginning of Chron C3n.4n. This interpretation relies on biostratigraphic data previously proposed for a part of the continental fauna found in the basin. This result is in agreement with the $^{40}Ar/^{39}Ar$ dating previously carried out on a volcanic ash layer, which provided an age of 5.9 ± 0.5 Ma [1] and which is shown here to be reversely magnetized. This layer is correlated here with the reverse polarity zone corresponding to Chron C3r.

Biostratigraphic studies on the same section have shown that the micromammal levels extend here only from middle Vallesian to upper Turolian (upper Miocene). Four localities have yielded western European species of micromammals, indicating trans-Mediterranean terrestrial faunal exchanges between these two continents during the late Miocene. The European murid rodent *Occitanomys* is recorded for the first time in North Africa in level 8 of the Afoud section, an age younger than 5.32 Ma being assigned to this level by the present study. Level 1 of the same section yields the lagomorph *Prolagus* cf. *michauxi*, with an age of 6.1 Ma.

The magnetostratigraphic data suggest therefore that the beginning of terrestrial faunal exchanges between North Africa and Western Europe took place in subchron C3An.1n, at about 6.1 Ma, some 0.4 Myr before the beginning of the Messinian salinity crisis.

Keywords: Morocco; upper Miocene; terrestrial environment; magnetostratigraphy; biostratigraphy; faunal provinces

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1. Introduction

The age of the micromammal faunas discovered in the Neogene continental formation of the Aït Kandoula basin extends from middle Miocene to upper Pliocene [1,2]. Along the Afoud section, which has a total thickness of about 290 m, a succession of six microvertebrate levels can be found. Four levels yield only a few fossils, but the fact that some of the rodent species are of western European origin suggests that some mammals crossed the marine straits separating the Moghreb from the Iberian Peninsula. This interpretation had been previously suggested by some authors [3–12]. Most of these authors have suggested that the interchange took place during the Messinian, but without being able to specify if it happened before, during or after the salinity crisis which characterizes this stage. In contrast, Aguilar et al. [7] suggested the interchange occurred before the salinity crisis. These authors consider that the Salobreña locality (Spain), which is contemporaneous to the Librilla and Venta del Moro deposits (Spain), belongs to the Neogene European mammal faunal zone MN13, but contains allochthonous species retaining primitive characters. Tortonian and Messinian paleogeographic reconstructions [13,14] for the western Mediterranean show no great differences between the two stages, with the existence of shoals at the North African side. Since this reconstitution, Aguilar et al. [7] concluded that shoals could have been used by terrestrial faunas as a passage between northern Africa and the Iberian Peninsula, following a change of sea level before the Messinian salinity crisis.

As magnetostratigraphy can be a useful tool for stratigraphic correlation in the continental domain, we chose the Afoud section for this study. Its continuous sedimentary succession, the presence of fossiliferous localities and the existence of an $^{40}Ar/^{39}Ar$ dated ash level (5.9 ± 0.5 Ma, [2]) could enable a correlation with the geomagnetic polarity time scale (GPTS). This correlation would allow the attribution of an absolute age for the micromammal fauna and the precise dating of the faunal exchange between North Africa and Western Europe during the Messinian.

2. Geological and biochronological setting

The Aït Kandoula basin is located on the southern slope of the High Atlas, northeast of Toundout village (Fig. 1). It has a surface area of approximately 330 km^2. The continental formations are mostly the product of the erosion. They accumulated in the Liassic dolomite syncline of the Toundout nappe [15,16]. There have been many controversies about the chronological position of these Neogene continental formations. They have been successively attributed to the Oligocene–Miocene by Roch [17], to the Pontian by Choubert [18] and to the Miocene–Pliocene by Gauthier [19]. Roch and Gauthier (op. cit.) based their dating on continental gastropods. Studying these same gastropods, Jodot [20] considered that the red formation of the Oued Tabia locality in the Aït Kandoula basin was of Pliocene age, but a first study of the rodent fauna of a fossiliferous level situated just above this formation yielded an early late Miocene age (middle Vallesian [21]). The first discovery of Neogene mammals in this basin was made by Jaeger [21] at the Oued Tabia locality. Afterwards, Görler and Zucht [2] discovered other fossiliferous localities in the same basin. To improve the biochronology of these continental formations, we resampled different parts of the Aït Kandoula basin. This led to the discovery of seven new fossiliferous localities. The mammalian fauna of these levels has already been studied in detail [1] allowing us to propose a middle Miocene (Astaracian) to upper Pliocene age for the Neogene Aït Kandoula formations (Fig. 2A)

In the western part of the basin, fluvio-lacustrine deposits consisting mainly of conglomerates, which attain in some cases a thickness of 40 m, attest to syn-orogenic activity. In the central part, an alternation of limestone, clay and ash, which corresponds to lacustrine or marshy sedimentation showing no visible discontinuity, can be recognized. This sedimentation corresponds to periods of relatively low tectonic activity. In the lower part of the Afoud section, which is situated some kilometres east of New Village (Fig. 1B), series of green, grey and red clay are predominant, while in its upper part an alternation of decimetric to metric limestone banks, marls and

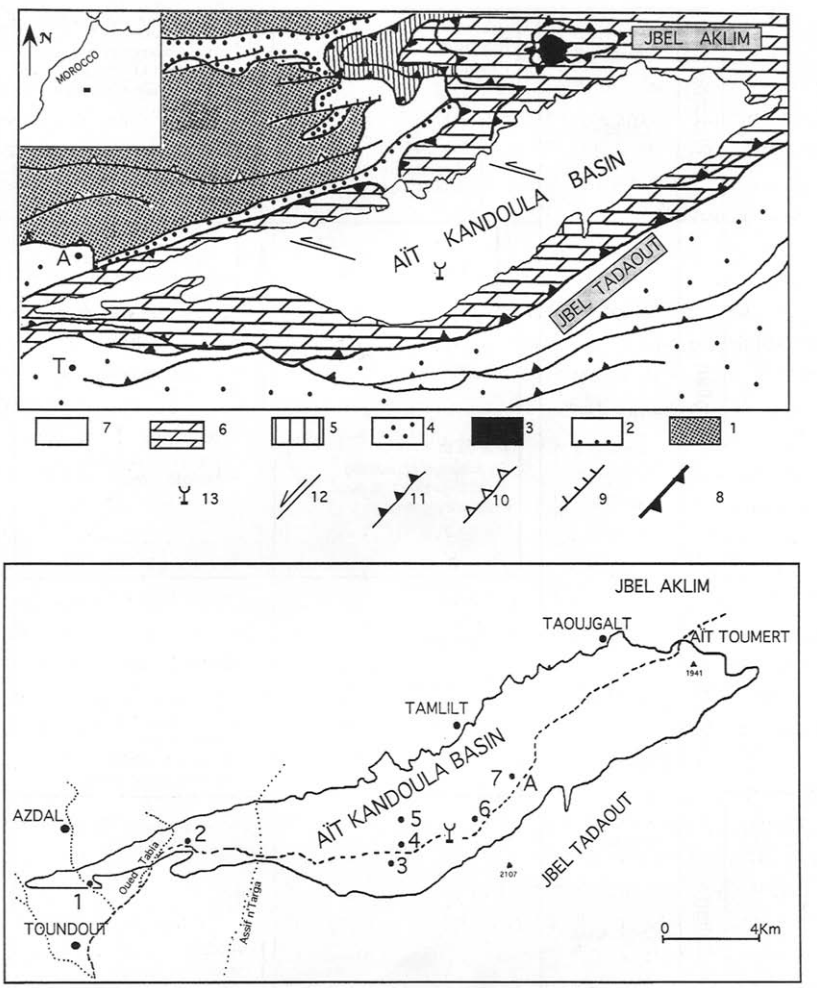
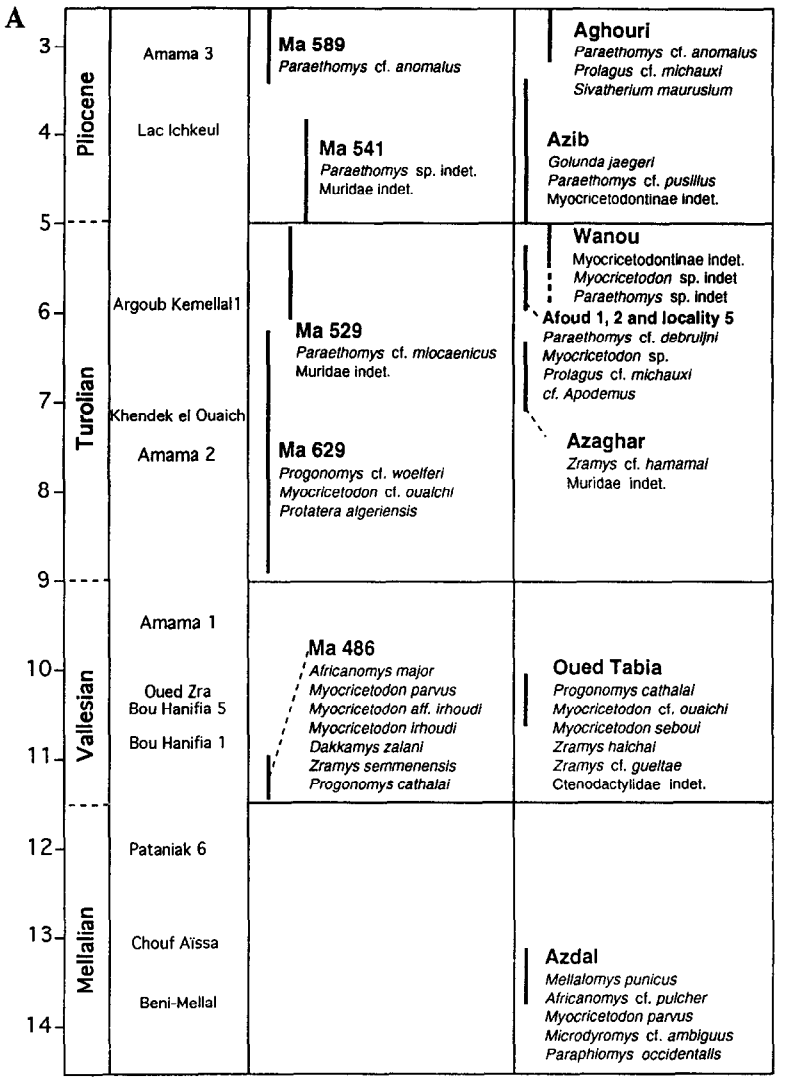


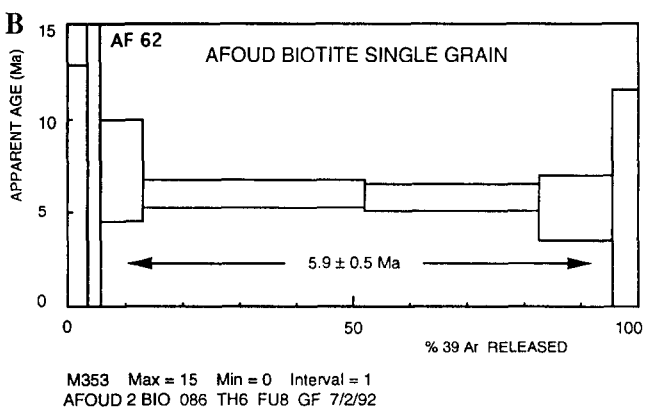
Fig. 1. Upper: Schematic geological map of the Aït Kandoula basin. *1* = Paleozoic substratum; *2* = Permo–Trias; *3* = Cretaceous windows; *4* = South Atlasic zone (Cretaceous–Eocene); *5* = para-autochtonous units; *6* = Liassic dolomite detachment; *7* = Neogene continental series; *8* = Permo–Trias/Toundout Liassic nappe contact; *9* = normal faults; *10* = reverse faults; *11* = minor thrusts; *12* = transverse faults; *13* = New Village; *T* = Toundout; *A* = Azdal (after Laville et al. [51]). Lower: Aït Kandoula Basin and situation of localities with micromammal findings. *1* = Azdal; *2* = Oued Tabia; *3* = Wanou; *4* = Azib; *5* = Aghouri; *6* = Azaghar; *7* = Afoud; *A* = Afoud section; ⅄ = New Village; - - - = track; ▲ = topographic heights.

clays can be recognized. Ash levels also appear in the upper part of the section, an ash seam yielding an $^{40}Ar/^{39}Ar$ age of 5.9 ± 0.5 Ma [1] (Fig. 2B).

3. Sampling and measurements

In a first session of fieldwork, 104 cores were taken at 68 levels from the Afoud section, which has a total thickness of 290 m. Most samples belonged to the upper part of this section (Fig. 3), where the average vertical sampling density is approximately one sampling level per 2.4 m. In the lower part, the marly clayish nature of the sediments did not allow such a thorough sampling. In most cases, two cores were taken from the same level. Some of the samples were collected using a gasoline-powered portable drill, but others were hand-size oriented blocks which were later drilled in the laboratory. Samples and blocks were oriented by means of a magnetic com-


A
3
4
5
6
7
8
9
10
11
12
13
14
Pliocene
Turolian
Vallesian
Mellalian
Amama 3
Lac Ichkeul
Argoub Kemellal 1
Khendek el Ouaich
Amama 2
Amama 1
Oued Zra
Bou Hanifia 5
Bou Hanifia 1
Pataniak 6
Chouf Aïssa
Beni-Mellal
Ma 589
Paraethomys cf. anomalus
Ma 541
Paraethomys sp. indet.
Muridae indet.
Ma 529
Paraethomys cf. miocaenicus
Muridae indet.
Ma 629
Progonomys cf. woelferi
Myocricetodon cf. ouaichi
Protatera algeriensis
Ma 486
Africanomys major
Myocricetodon parvus
Myocricetodon aff. irhoudi
Myocricetodon irhoudi
Dakkamys zalani
Zramys semmenensis
Progonomys cathalai
Aghouri
Paraethomys cf. anomalus
Prolagus cf. michauxi
Sivatherium maurusium
Azib
Golunda jaegeri
Paraethomys cf. pusillus
Myocricetodontinae indet.
Wanou
Myocricetodontinae indet.
Myocricetodon sp. indet
Paraethomys sp. indet
Afoud 1, 2 and locality 5
Paraethomys cf. debruijni
Myocricetodon sp.
Prolagus cf. michauxi
cf. Apodemus
Azaghar
Zramys cf. hamamai
Muridae indet.
Oued Tabia
Progonomys cathalai
Myocricetodon cf. ouaichi
Myocricetodon seboui
Zramys haichai
Zramys cf. gueltae
Ctenodactylidae indet.
Azdal
Mellalomys punicus
Africanomys cf. pulcher
Myocricetodon parvus
Microdyromys cf. ambiguus
Paraphiomys occidentalis


B
AF 62
AFOUD BIOTITE SINGLE GRAIN
APPARENT AGE (Ma)
15
10
5
0
5.9 ± 0.5 Ma
0
50
100
% 39 Ar RELEASED
M353 Max = 15 Min = 0 Interval = 1
AFOUD 2 BIO 086 TH6 FU8 GF 7/2/92

pass. At the laboratory, two to four 2.2 cm specimens were cut from each core.

The paleomagnetic analysis of 113 specimens was carried out at the paleomagnetic laboratory of the University of Montpellier II. Remanent magnetization was measured using a 2 axis CTF cryogenic magnetometer. These measurements were recorded after stabilization of remanence in the cryogenic magnetometer, which was made possible by monitoring magnetization changes in real time [22]. Most of the specimens were subjected to stepwise thermal demagnetization up to a maximum temperature of 670°C, while 11 were subjected to stepwise alternating field demagnetization up to maximum fields of 35–200 mT. For a few samples a combined treatment of thermal demagnetization up to 150°C and subsequent AF demagnetization was chosen. Thermal demagnetization steps were selected between 20 and 50°C and alternating field demagnetization steps between 5 and 10 mT. During thermal demagnetization, initial susceptibility at room temperature was measured after each step, to check if chemical and/or mineralogical changes had taken place in magnetic minerals during heating. Orthogonal vector plots and stereographic projections were used for analysis of remanence directions and polarities. Directions of magnetization components were calculated by means of principal component analysis [23].

In addition, isothermal remanent magnetization (IRM) experiments were carried out on 7 representative samples of the Afoud section. These experiments consisted of stepwise acquisition of IRM with an electromagnet up to a maximum field of 0.8 T and subsequent thermal demagnetization, with susceptibility at room temperature being measured after each step.

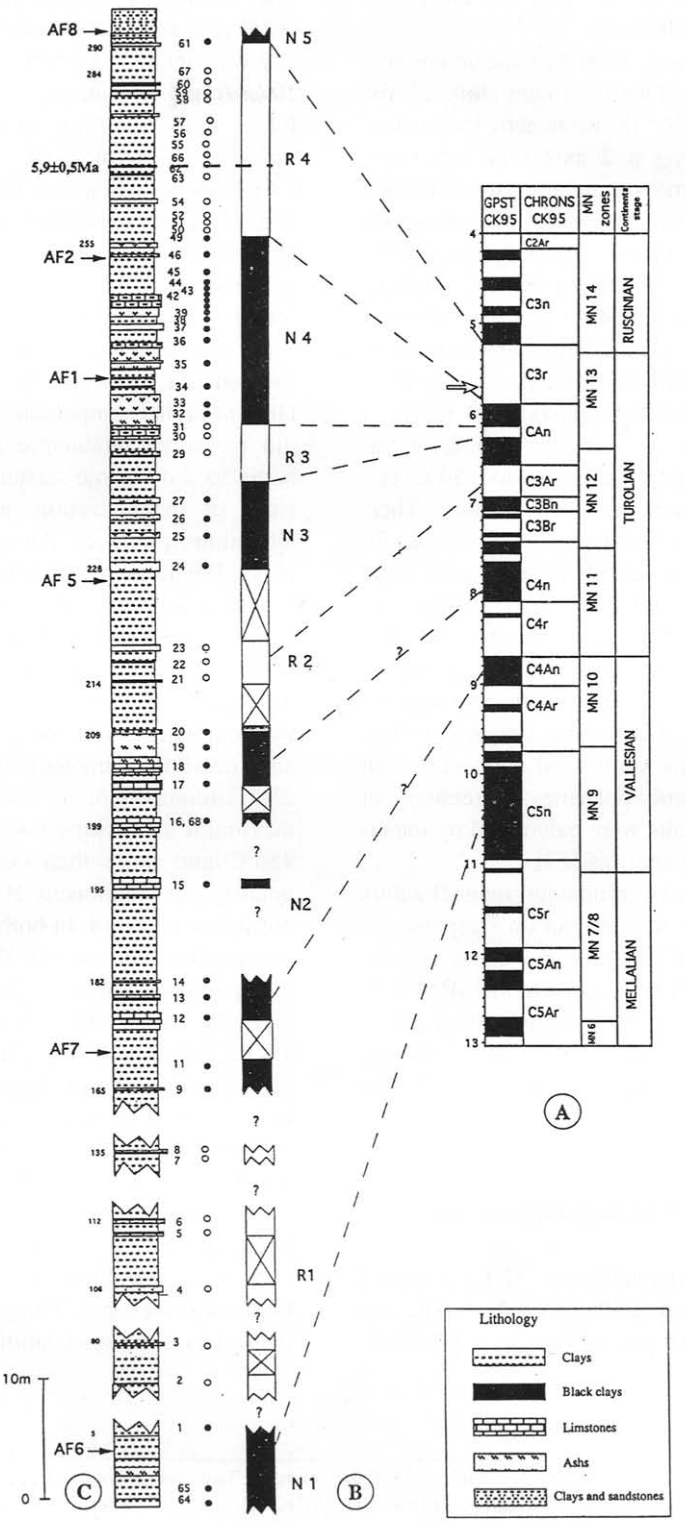
4. Paleomagnetic and rockmagnetic results

Natural remanent magnetization (NRM) varied between 2.9×10^{-9} and 2.3×10^{-5} Am^2/kg, this wide range of variation being a function of lithology. The lithologies sampled included limestones, clays, marls and volcanic ashes. The weakest NRM intensity was obtained for a highly altered cinerite (2.9×10^{-9} Am^2/kg) while the highest values were found for a red clay layer at the base of the section (4.1×10^{-6} Am^2/kg), the already mentioned $^{40}Ar/^{39}Ar$ dated cinerite layer (1.1×10^{-5} Am^2/kg) and a clayish sandstone layer at the top of the section (2.3×10^{-5} Am^2/kg). Susceptibility values, varying between 2.6×10^{-10} and 8.4×10^{-7} m^3/kg, were found to be generally correlated to NRM intensities.

Of the thermally demagnetized specimens, 60% showed stable behaviour during demagnetization. Only at higher temperatures — more than 500°C — did measured remanence directions in some cases begin to show some scattering, caused by low intensities of magnetization or chemical/mineralogical alterations produced during heating. In other samples, directions remained stable until the highest temperatures, sometimes more than 600°C. Besides a small and easily removed laboratory-acquired viscous component, two main components could be distinguished in all samples (Fig. 4): a normal polarity component with the present-day field direction and maximum unblocking temperatures of 200–250°C (component S) and another component with maximum unblocking temperatures varying between 450°C and more than 600°C (component P). The polarity of component P was found to be either normal or reversed. In both cases, its declination was not significantly rotated. Both components could be easily distinguished from each other in those samples in which component P had a reversed polarity (Fig. 4A,B), but they were often difficult to separate when the polarity of P was normal (Fig. 4D). The reason for this difficulty was the relatively low dips of bedding planes ($< 20°$), so that in some cases both S and P directions were very similar.

The behaviour of 41% of the thermally demagnetized specimens was unstable during demagnetization: at temperatures over 300°C remanence directions were scattered. This behaviour is due to the low intensities of magnetization of these samples and/or

Fig. 2. (A) Stratigraphic range chart of rodent fauna in the Aït Kandoula basin. Left: after Görler and Zucht [2]; to the right after Benammi et al. [1]. (B) $^{40}Ar/^{39}Ar$ age spectra for biotite samples of ash level AF 62 [1].


AF8
AF2
AF1
AF 5
AF7
AF6
5,9±0,5Ma
10m
0
N 5
R 4
N 4
R 3
N 3
R 2
N2
R1
N 1
GPST CK95
CHRONS CK95
MN zones
Continental stage
C2Ar
C3n
C3r
CAn
C3Ar
C3Bn
C3Br
C4n
C4r
C4An
C4Ar
C5n
C5r
C5An
C5Ar
MN 14
MN 13
MN 12
MN 11
MN 10
MN 9
MN 7/8
MN 6
RUSCINIAN
TUROLIAN
VALLESIAN
MELLALIAN
A
B
C
Lithology
Clays
Black clays
Limstones
Ashs
Clays and sandstones

chemical/mineralogical alterations upon heating, as shown by changes of susceptibility. Often an important initial fraction of the total remanence — a viscous or chemical remanent remagnetization — was destroyed at temperatures below 150°C. Nevertheless, although it was not possible to determine the primary paleomagnetic directions, often the magnetic polarities could be recognized (Fig. 4C). The trend of remagnetization circles was very useful for determining the polarity of the primary component when it was reversed (Fig. 4F). In most cases, susceptibility showed a slight increase at temperatures over 200°C and a stronger increase over 400°C. In several cases a stable behaviour could also be observed.

Only 11 specimens were submitted to AF demagnetization. Their behaviour during demagnetization was not homogeneous. Several samples could only be demagnetized up to a maximum field of 35 mT. At higher fields incoherent directions were measured. Component S could be well determined, but only the polarity of component P could be recognized. Other samples could not be completely demagnetized, even after application of higher fields up to 200 mT. Strong deviations from the directions observed in most other samples were obtained. Thermal demagnetization of specimens of the same cores showed that this behaviour was probably due to the simultaneous presence of two higher coercivity phases. Nevertheless, some samples subjected to a composed thermal and AF treatment yielded good results (Fig. 4B).

IRM acquisition curves show the presence of mainly low-coercivity minerals. IRM values increase quickly and with relatively weak fields of 0.1–0.2 T more than 90% of saturation remanence seems to be reached (Fig. 5A). To saturate samples, fields stronger than 0.5 T were needed. In all demagnetization curves maximum unblocking temperatures were lower than 590°C (Fig. 5B). Magnetite as a carrier of remanence seems to be present in all six samples studied. Also, a phase of low unblocking temperatures can be recognized.

5. Discussion

5.1. Correlation

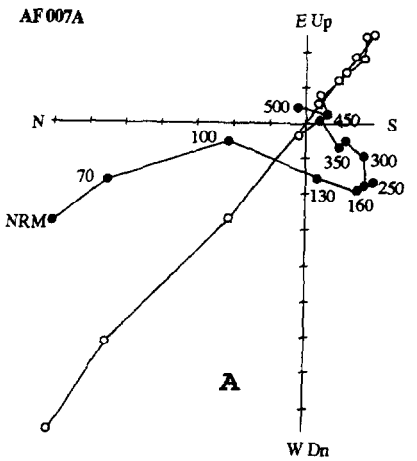
In 60 of the 113 studied specimens a direction for component P could be determined. In 8 cases, two specimens belonged to the same core or stratigraphic level, so that directions for 52 levels (36 normal and 16 reversed) were obtained. The mean direction for normal-polarity samples was D = 349.4, I = 50.7 (N = 36, $\alpha_{95} = 4.5$, k = 27) and for reversed polarity samples D = 191.2, I = −53.4 (N = 16, $\alpha_{95} = 12.32$, k = 9). Both directions are not significantly different, so that they can be considered as "roughly" antipodal. In 53 specimens at least the polarity of component P could be recognized.

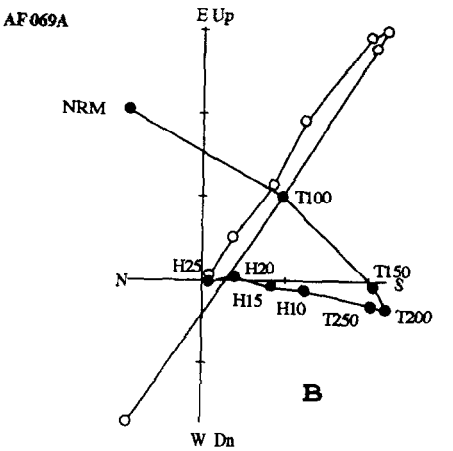
We consider component P a primary component for the following reasons:

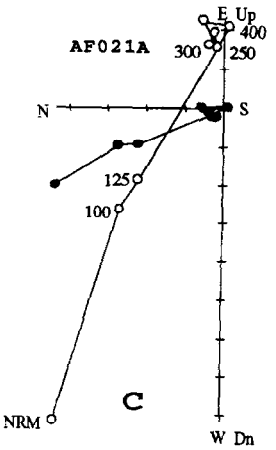
1. two components, S and P, can be normally recognized: S is a present-day normal polarity component and is different from P;
2. normal and reversed polarities are recognized, thus excluding remagnetization of the section;
3. normal and reversed component P directions are "roughly" antipodal (Fig. 6).

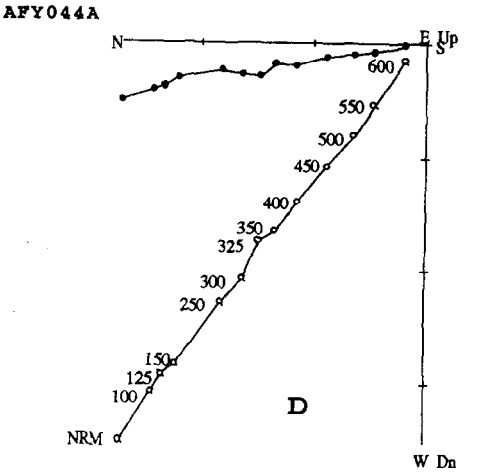
Paleomagnetic studies on continental series are scarce, at least for the Neogene. A correlation of the European continental stages with the geomagnetic polarity time scale of Berggren et al. [24] has been proposed by Steininger et al. [25], who, for instance, correlate the Turolian (mammal faunal units MN11, MN12 and MN13) with polarity Chron C3r to

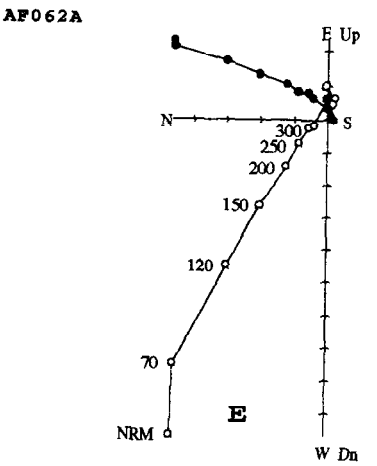
Fig. 3. Lithology, magnetostratigraphy and stratigraphic position of sampling sites of the Afoud section. Correlation of the Afoud section to the GPTS of Cande and Kent [27] (see text). *Column A*: Segment of the magnetic polarity time scale of Cande and Kent [27]. Right-hand columns display mammalian zones (Mein, 1990 [52]) and European continental stages (Mellalian = Astaracian, a continental stage in the Moghreb, Jaeger et al. [4]). Limits of mammal faunal units after Sen [31]; Opdyke et al. [28]; Sen et al. [29]; Krijgsman et al. [30]; and Garcés et al. [33]. Arrows point to position of the beginning of the Messinian salinity crisis (Gautier et al., [48]). Question marks denote uncertain correlation. *Column B*: N1–N5 = normal polarity intervals; R1–R4 = reversed polarity intervals, crossed area in the polarity column denotes a non 'sampling' lithology. *Column C*: Lithology displays variations in coloured clays, limestones, ash and sandstones. *AF 6, AF 7, AF 5, AF 1, AF 2, and AF 8* = localities with micromammals. The numbers at the left side indicate the stratigraphic levels.

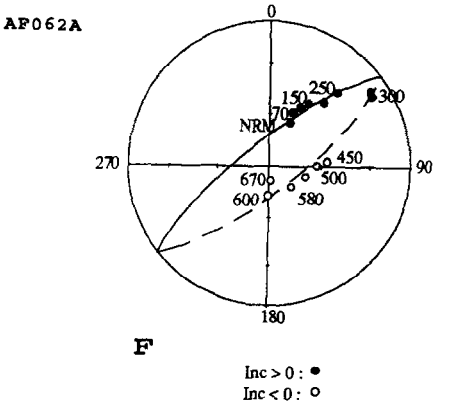

AF007A
E Up
N
S
500
450
100
350
300
70
250
130
160
NRM
W Dn
A


AF069A
E Up
NRM
T100
H25
H20
T150
N
S
H15
H10
T250
T200
W Dn
B


AF021A
E Up
400
300
250
N
S
125
100
NRM
W Dn
C


AFY044A
N
E Up
S
600
550
500
450
400
350
325
300
250
150
125
100
NRM
W Dn
D


AF062A
E Up
N
S
300
250
200
150
120
70
NRM
W Dn
E


AF062A
0
150
250
300
70
NRM
270
90
450
670
500
600
580
180
F
Inc > 0 : •
Inc < 0 : ○

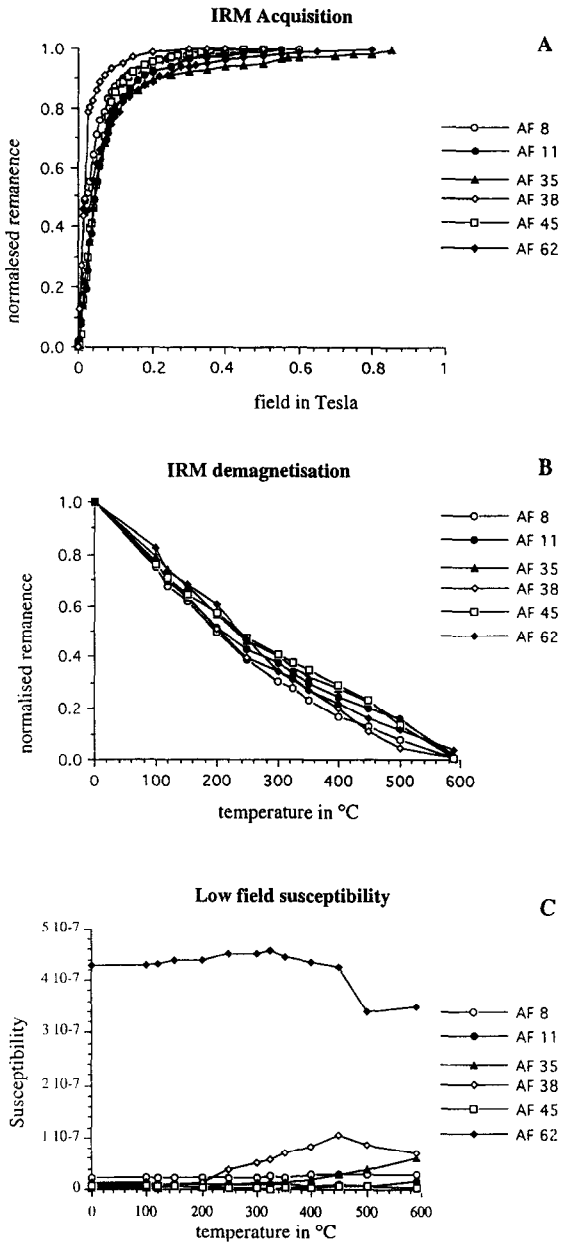


Fig. 5. (A) Isothermal remanent magnetization (IRM) acquisition curves of selected samples. (B) Thermal demagnetization of IRM acquired at 0.8 T. (C) Susceptibility values after each demagnetization step measured at room temperature for selected samples of the Afoud section.

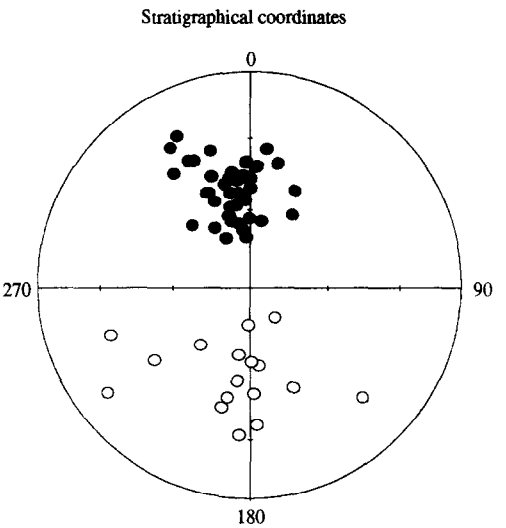


Fig. 6. Stereographic projection of normal and reversed primary components.

C4Ar.3. The magnetostratigraphic analysis of a section in Samos, Greece [26], places the MN12–MN13 stage limit at the base of Chron C3An.2n, (6.56 Ma, following the new polarity time scale of Cande and Kent (CK95) [27]. At the Cabriel basin (Spain), the MN11–MN12 transition is correlated with Chron C4n.1n, at 7.2–7.3 Ma [28]. The same conclusion was reached for the Kemiklitepe section in Turkey [29]. In the GPTS of Cande and Kent [27], this transition is placed between 7.7 and 8.0 Ma. A magnetostratigraphic study carried out recently by Krijgsman et al. [30] in several basins from eastern Spain yielded an age of 8.7 ± 0.1 Ma for the lower limit of MN11 and 7.5 ± 0.1 Ma for the MN11–MN12 boundary.

For a long time, the appearance of the *Hipparion* genus fixed the age of the Vallesian lower limit. Magnetostratigraphic correlation of the Bou Hanifia section in Algeria [31] and Kastellios Hill in Greece [32] showed that, around the Mediterranean, the Vallesian lower limit was not older than 11.5 Ma. A high resolution magnetostratigraphic study carried

Fig. 4. Demagnetization vector plots and stereographic projections of selected samples from the Afoud section. Plots A, C–F show results of stepwise thermal demagnetization and plot B shows a combined alternating field (H) and thermal (T) demagnetization. ● = the horizontal projection of the NRM vector end-points during demagnetization; ○ = vertical projection of the NRM vector end-points. Demagnetization temperatures are given in °C and fields in mT.

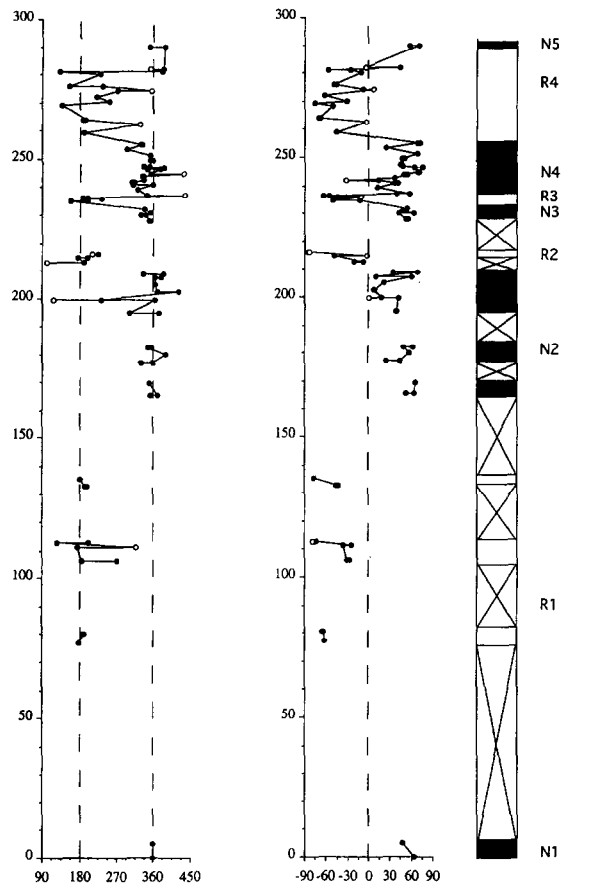


Fig. 7. Inclination and declination of ChRM as a function of stratigraphical level. ○ = low intensity samples which are difficult to interpret. In the polarity column, black denotes normal polarity and white denote reversed polarity zones. Crossed area in the polarity column denotes a non 'sampling' lithology.

out in the Vallés–Penedés basin (NE Spain) by Garcés et al. [33] sets the lower limit of the Vallesian stage at 11.1 Ma. This age agrees with the radiometric ages of other localities around the Mediterranean. Previously, the Vallesian upper limit was set at 9 Ma by Steininger et al. [25], while recently an age of 8.7 ± 0.1 Ma was found by Garcés et al. [33] for this limit in the Duero basin (Spain).

Figs. 3 and 7 show the magnetostratigraphy obtained in the present study for the Afoud section; six normal and four reversed polarity zones can be distinguished. Several sampling gaps occur between these ten polarity intervals, separating in some cases zones of the same polarity and in other cases zones of different polarity. Most likely, more than 9 polarity zones are present. Gaps occur mainly in the lower part of the section, and are generally due to the unsuitability of some lithologies for sampling.

Correlation of the polarity succession obtained in the present study with the geomagnetic polarity time scale of Cande and Kent [27] has been performed taking into consideration the following facts: (1) the mammals found in these deposits place this section in the late Miocene; (2) the mammals from Afoud 1, 2, 5 and 8 belong to zone MN13; (3) the degree of evolution of a faunal association found in Afoud 6 permits placement in the Vallesian (about 8.7–11.1 Ma), close to the MN9–MN10 boundary (9.7 Ma, Krijgsman et al. [30,33]); and (4) $^{40}Ar/^{39}Ar$ dating of an ash layer in the upper part of the section yields an age of 5.9 ± 0.5 Ma.

Clearly, for the upper part of the Afoud section, unambiguous correlation can be proposed. This part is dated as late Turolian. The fauna found in levels Afoud 1, 2, 5, and 8 (with genus: *Apodemus*, *Prolagus*, *Occitanomys* and *Cricetus*) belongs to MN13; all these genera are of western European origin. These mammals sites belong to distinctive magnetozones. Afoud 5 belongs to the base of N3 and can be correlated to Chron CAn.2n; R3 can be correlated to Chron C3An.1r, therefore Afoud 1 and 2, situated in magnetozone N4, must be correlated to Chron C3An.1n (Fig. 3). R4 is correlated to Chron C3r, and Afoud 8, which is located in zone N5, and which is based on a single level sampling, is correlated to Chron C3r/C3An.4n boundary. This correlation is in agreement with the $^{40}Ar/^{39}Ar$ age of ash level from zone R4 which is correlated to Chron C3r.

In the lower part of the section, Afoud 6 shows a faunal association whose degree of evolution lies between that of the fauna from Bou Hanifia 5 (Algeria) and that of the fauna from Oued Zra (Morocco). The Oued Zra deposit has a radiometric age of 9.7 ± 0.5 Ma [21]. A magnetostratigraphic study carried out in Bou Hanifia [31] yielded an age of 10.3 Ma. for Bou Hanifia 5 after correlation with the polarity time scale of Harland et al. [34]. A revision of this result with the GTPS of Cande and Kent (1995) yields an age of 10.9 Ma. Sen [31] had based his correlation on the ash level dated at 12.03 ± 0.25 Ma. This level is situated about 165 m below that of the fossil locality of Bou Hanifia 5. Since there are

large stratigraphic gaps between the ash layer and Bou Hanifia 5, this locality would be correlated with Chron C4Ar (Sen, pers. commun.), which is suggested by Garcés et al. [33] and Krijgsman et al. [30]. Afoud 6 could thus have an age older than 10.1 Ma, which is corroborated by the existence of micromammals of lower Vallesian age at site 486 [2], which lies some metres below Afoud 6 in the same section.

As the correlation is in agreement with the radiometric dating and the magnetostratigraphic carried out in the other continental formations [26,28–30,33], we suggest that the Afoud section covers a continuous time interval of almost 5.18 Myr. The base of the section is most likely situated at Chron C5n.2n and has an age older than 10.1 Ma, the top lies in Chron C3r/C3n.4n boundary and has an age of 5.23 Ma, according to CK95. The different lithologies of the Afoud section suggest changing sedimentation rates. For the upper part, a mean sedimentation rate of about 6 cm/kyr is obtained (214–290 m), and about 5 cm/kyr for the lower part (0–214 m). This difference in sedimentation rates can be explained by the dominance of volcanic ash levels in the upper part of the section. Our correlations allow us to assign the following approximate ages to the fossiliferous deposits of the Afoud section (Fig. 8): 10.5–10.1 Ma for Afoud 6; an age younger than 7.5 Ma for Afoud 7; an age older than 6.3 for Afoud 5; an age of ~ 6.1 Ma for Afoud 1; an age younger than 5.8 Ma for Afoud 2; and an age younger than 5.23 Ma for Afoud 8.

5.2. *Salinity crisis and age of faunal exchanges between the Moghreb and southwestern Europe*

The salinity crisis which took place in the Mediterranean at the Messinian was one of the strongest environmental changes geologically recorded during the Tertiary. Since the discovery of evaporites in the Mediterranean [35] several studies dealing with the upper and lower age limits of the Messinian stage have been published. D'Onofrio et al. [36] proposed that the lower boundary of the Messinian be defined by the first appearance of *Globorotalia conomiozea* in the Falconara section in Sicily, which is presently accepted as a convention. The paleomagnetic study of the same section has not

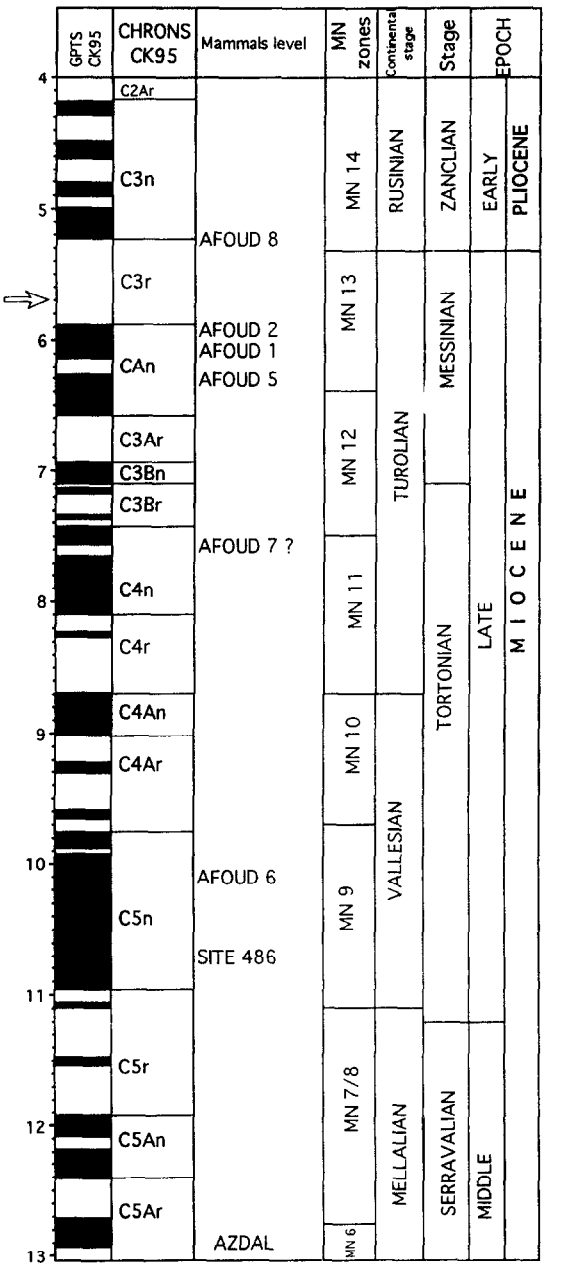


Fig. 8. Stratigraphic position of mammal localities from Afoud section and proposed correlation with the geomagnetic polarity time scale of Cande and Kent [27]. Arrows point to position of the beginning of the Messinian salinity crisis [48]. Limits of mammal faunal units after Sen [31]; Opdyke et al. [28]; Sen et al. [29]; Krijgsman et al. [30] and Garcés et al. [33].

allowed Langereis and Dekkers [37] to place the Tortonian–Messinian boundary because the section has been remagnetized after tilting. In another paleomagnetic study on other sections (central and eastern Crete) Krigjsman et al. [38] correlated the Tortonian/Messinian boundary with subchron C3Br.1r (7.10 Ma). On Gavdos (Greece) and on Sicily, the first occurrence of *Globorotalia conomiozea* (Tortonian–Messinian boundary) was determined at Chron C3Br.1r and had an age of 7.12 Ma [39], according to CK95, and an astronomical age of 7.24 Ma (or 7.21 Ma [40]).

In the Messinian stratotype in Sicily, the zone corresponding to the salinity crisis is not characterized by any marine fossil, which makes correlation with contemporaneous sections situated outside of the Mediterranean region difficult. Benson and Rakic-El Bied [41] succeeded in correlating the late Miocene Bou Regreg section near Rabat (Morocco) with the magnetic polarity time scale, thus allowing the calibration of Messinian fauna which had not been correlated up to that time. This section is well studied from a magnetostratigraphic, biostratigraphic and stable isotopic point of view [42–45]. The first appearance of *G. conomiozea* was placed in the Bou Regreg section at Chron C3Bn [41,44].

The Miocene/Pliocene boundary (the base of the Zanclian stage) reflects the re-establishment of open-marine conditions after the late Miocene isolation of the Mediterranean. This boundary is defined by the base of the Trubi on Sicily, and the astronomical age of this boundary is dated at 5.32 Ma (early Gilbert) [45,46]. In contrast, Krijgsman et al. [39] have dated this boundary at 5.33 Ma. This means that the Messinian stage lasted approximately 1.80 or 1.91 Myr (with the recent astronomical calibration of Hilgen et al. [40]). In our study we take a duration of 1.8 Myr for this stage.

The progressive closure of the South Rifian Strait is recorded at the Bou Regreg section and included in subchron C3An.1n (6.1–5.8 Ma). This dating is based on the last appearance of normal marine plankton and the first change in the coiling direction of *Neogloboquadrina acostaensis* [47].

Gautier et al. [48] carried out a magnetostratigraphic study on sections in Sorbas (southern Spain) and the Caltanisseta basin (Sicily) trying to solve the Messinian salinity crisis problem. They suggested that the salinity crisis covered most of the C3r Chron (i.e. from ~ 5.7 to 5.3 Ma).

Some upper Turolian (Messinian) fossiliferous localities in northern Africa yield micromammal species showing affinities with southwestern European species. These localities include Aïn Guettara in Morocco [5], La Voie Ferrée in Tunisia [49], Argoub Kemellal 1 in Algeria [12] and Afoud (1, 2, 5 and 8) and Aghouri in Morocco [1]. In the present study, correlation of the Afoud section with the GPTS of Cande and Kent [27] suggests that the oldest fossiliferous level that yields micromammals with European affinities (Afoud 1, Fig. 3) is situated 3 m above the base of Chron C3An.1n. These 3 m correspond to a sedimentation time interval of 40 kyr, as has been calculated from the mean sedimentation rate of polarity interval N 4 (7.5 cm/kyr). Therefore, Afoud 1 would have an age of about 6.1 Ma, which suggests that faunal exchanges had taken place before the starting of the salinity crisis. The same could be concluded even if a mean sedimentation rate of 6 cm/kyr for the whole Afoud section was taken. The second fossiliferous deposit, Afoud 2, has an age older than 5.8 Ma. It should be noticed that in another deposit (Afoud 5) a molar of an European genus, *Apodemus*, has been found. This locality is situated several metres below Afoud 1 and exposed at some metres of lateral distance from that deposit.

At the Tortonian/Messinian boundary, syn-sedimentary tectonic activity could have controlled the level of sills, particularly at the South Rifian strait, and could have reduced and locally cut off communications. These exchanges were possibly a consequence of tectonic movements, glacio-eustatic events and particular local conditions [4,41,50]. The Betic-Rif area could have been the way for the faunal exchange between North Africa and southwestern Europe at the end of the Miocene.

6. Conclusion

The present magnetostratigraphic study of the Afoud section shows that a minimum of ten polarity zones can be recognized. Following this assumption, we suggest that this section was deposited during the time interval between subchron C5n.2n and the be-

ginning of subchron C3n.4n. The corresponding duration is about 5.18 Myr. The age of a volcanic ash layer situated in the upper part of the section in the reverse magnetozone R 4 is correlated to Chron C3r, this result corroborated our magnetostratigraphic correlation with the GPTS. The faunal association of the middle Vallesian age found at the Afoud 6 level, approximately dates the lower part of the section and suggests the correlation of polarity zone N1 to Chron C5n.2n. From a geodynamic point of view, the infilling of the Aït Kandoula basin appears to correspond to a regular sedimentation rate, at least from Oued Tabia to the oriental part of the basin.

The present study has allowed us to attribute approximate numerical ages to the fossiliferous levels of the same section. Consequently, the beginning of the terrestrial fauna exchanges between North Africa and southwestern Europe, as registered in this basin, would have occurred before the Messinian salinity crisis. This faunal interchange would have happened at the moment of closure of the South Rifian strait, during subchron C3An.1n at 6.1 Ma.

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